Ganesh Malepati

malepatiganesh3@gmail.com 8328100611 in/ganesh-malepati-34855a1b8

SUMMARY

SDET with 2 years of experience building and scaling UI and API automation using Python, Pytest, and Selenium. Experienced in designing reliable test frameworks, reducing flaky test, and integrating automation into CI/CD pipelines for continuous delivery. Hands-on exposure to AI-driven systems, including validation of LLM-based diagnosis pipelines and anomaly detection models using RAG. Strong focus on improving system reliability, debugging complex failures, and ensuring production-grade quality across distributed systems.

EXPERIENCE

Capgemini **May 2024 - Present**

- Designed and maintained UI & API automation test cases using Pytest, Selenium, and REST APIs, improving regression coverage for critical modules.
- Reduced test flakiness by 30% by refactoring selenium scripts with better synchronization and element handling techniques.
- Contributed to the development of an Agentic AI-based Auto-Heal system for anomaly detection, incident correlation, and automated diagnosis using multi-sensor data.
- Validated and compared SLM models (LLaMA 3.2, Phi-3) with RAG-based grounding for generating actionable diagnosis outputs. Ensured end-to end pipeline reliability, performance (SLOs), and consistency across CLI and UI-based workflows.

PROJECTS

PROJECT – AutoHeal On-Box **January 2026 - April 2026**

- Performed data analysis, validation, and testing across datasets (Ordered Mixed Data, CaseData, Tag-UseCase) to ensure data integrity and correctness.
- Validated anomaly detection models using F1-score, Recall, Event Recall, and Detection Delay metrics, ensuring ≥90% accuracy on eligible sensors. Verified multi-sensor correlation logic (e.g., Battery + Fan, Memory + Thermal) and validated incident outputs across pipeline stages.
- Tested with Small Language Models (LLaMA 3.2 vs Phi-3) for diagnosis generation, ensuring structured outputs (intent, steps, verify, risk) and KB grounded responses using RAG. Tested end-to-end pipelines (detect → correlate → diagnose) and ensured consistency between CLI and UI outputs. Analyzed debug traces and telemetry logs to improve pipeline reliability and success rate.

PROJECT – Nutanix Cloud Management **June 2024 - December 2025**

- Automated 20+ UI and API test cases using Pytest and Selenium, achieving 30% regression coverage for critical modules.
- Reduced false positives by 30% by refactoring legacy Selenium scripts and stabilizing flaky test cases through improved synchronization and element handling.
- Integrated automated suites into Jenkins pipelines, enabling nightly CI runs and continuous feedback for development teams.
- Added new test functionalities and utilities to the existing automation framework based on evolving client requirements, improving adaptability and scalability of the test suite. Maintained automation framework enhancements, adding utilities for data-driven testing and reporting using pytest plugins and HTML reports.

SKILLS

Programming & Scripting: Python (Proficient)
Test Automation Frameworks: PyTest, Selenium, RESTful
API Tools: Postman, JSON / REST-ful Services
DevOps: GitHub, Fundamentals of Jenkins and Docker
Environment IDE: Pycharm, VScode and Linux Commands
Data & AI/ML: ML-Models, Deep Learning, NLP, LLMs & SLMs (Foundational Knowledge)
Libraries: Pandas, NumPy, Py-Torch (Basic), Tensor-Flow (Basic)
Databases: My-SQL(Basic Queries)
Cloud Platforms: AWS (Cloud Practitioner)
Core Concepts: OOPs, SDLC/STLC, Defect Life Cycle
Key Strengths: Strong debugging, clean coding practices, collaborative mindset, Agility to learn new technologies.

EDUCATION

Bachelor of Technology — Computer Science and Engineering

CMR Engineering College, Hyderabad — 2023 — 70%

CERTIFICATIONS

- AWS Cloud Practitioner — (AWS, 2025)
- Python — (Hacker-Rank, 2024)
- Smart Coder — (Smart Interviews, 2022)